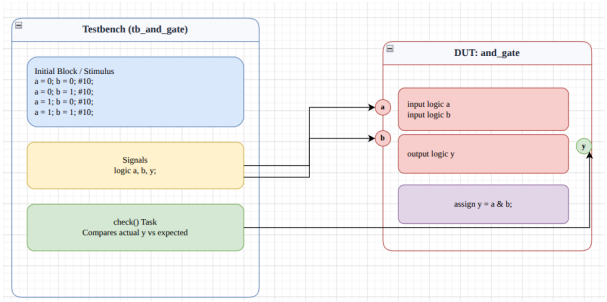
MEDS: Module 2

Zain Saeed
Roll No: 2026msee25

Task Name: Task 5

Git Repository: https://github.com/ZaynSaeed04/MEDS-MODULE-2.git

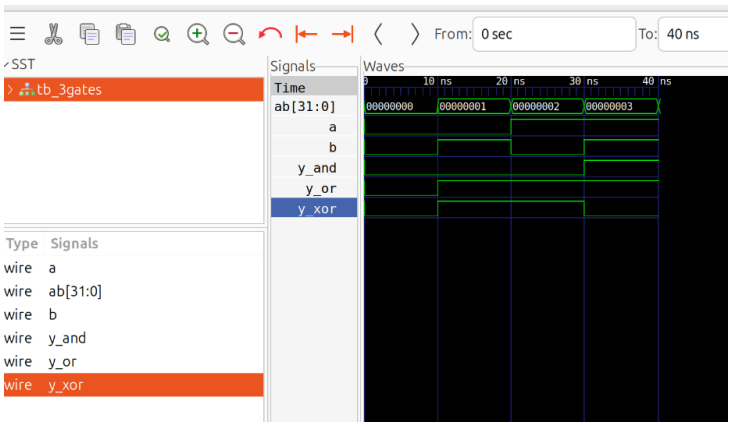
Schematic



Terminal Output

```
tes.sv
- Verilator: Built from 0.032 MB sources in 5 modules, into 0.054 MB in 11 C++ files need
- Verilator: Walltime 8.624 s (elab=0.001, cvt=0.011, bld=8.603); cpu 0.035 s on 12 threa
zain-saeed@zain-saeed-Lenovo-V14-G3-IAP:~/Desktop/MEDS/M2/T5$ ./obj_dir/Vtb_3gates
[PASS] AND ab=00: value=0 at time=10 ns
[PASS] OR ab=00: value=0 at time=10 ns
[PASS] XOR ab=00: value=0 at time=10 ns
[PASS] AND ab=01: value=0 at time=20 ns
[PASS] OR ab=01: value=1 at time=20 ns
[PASS] XOR ab=01: value=1 at time=20 ns
[PASS] AND ab=10: value=0 at time=30 ns
[PASS] OR ab=10: value=1 at time=30 ns
[PASS] XOR ab=10: value=1 at time=30 ns
[PASS] AND ab=11: value=1 at time=40 ns
[PASS] OR ab=11: value=1 at time=40 ns
[PASS] XOR ab=11: value=0 at time=40 ns
All tests completed.
```

Waveform



Explanation

In this task, I implemented and verified combinational logic for an AND gate alongside an extended multi-gate architecture (`3gates.sv`). The attached schematic represents the basic `and_gate`, while the schematic for `3gates` follows an identical structural layout with expanded gate instantiations. The included terminal logs and GTKWave waveforms correspond directly to the comprehensive multi-gate testbench (`tb_3gates.sv`). A custom `check()` task using 4-state inequality (`!==`) automatically validated all input combinations, logging explicit `[PASS]` or `[FAIL]` statuses with exact timestamps.